THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** accommodation	Ciliary muscles
2	**(a)** 25 cm	Near point
3	**(b)** concave lens	Diverging
4	**(a)** Correct	Fine particles
5	**(b)** Student A is correct	Violet deviates most
6	**(a)** dispersion	Spectrum
7	**(c)** atmospheric refraction	Changing refractive index
8	**(c)** iris	Pupil size
9	**(c)** is scattered the least by fog/smoke	Longest wavelength
10	**(b)** hypermetropia	Far-sightedness

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Far point: infinity; near point: about 25 cm.	1+1
12	Stars are point sources, so atmospheric refraction changes their apparent brightness; planets are extended sources whose variations average out.	2
13	Loss of accommodation with age, due to weakening of ciliary muscles and reduced flexibility of eye lens.	2
14	True — there is no atmosphere to scatter light.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Near-sightedness — distant objects are not clear; causes: excessive curvature of eye lens, elongation of eyeball; corrected by a concave lens of suitable power	1+1+1
16	Light travels a long distance through the atmosphere near the horizon; most blue light is scattered away; red light (longer wavelength) reaches the eye	3
17	Ray bends towards the base on entering and leaving; angle between the incident ray produced and emergent ray is the angle of deviation	3
	OR: Natural spectrum after rain; water droplets act as prisms: refraction and dispersion on entering, internal reflection, refraction on leaving	1+2
18	Far-sightedness — near point farther than 25 cm; causes: focal length of eye lens too long, eyeball too small; corrected by convex lens	1+1+1
	OR: Due to atmospheric refraction, the Sun is visible about 2 minutes before actual sunrise and after actual sunset	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) cornea, iris, pupil, eye lens, ciliary muscles, retina, optic nerve (ii) retina: light-sensitive screen; cornea: most refraction; ciliary muscles: change lens curvature; optic nerve: carries signals to brain	2+3
	OR: (i) splitting of white light into colours (ii) different colours bend through different angles (refractive index differs) (iii) second inverted prism recombines the colours to give white light (iv) band of coloured components VIBGYOR	1+1+2+1
20	(i) myopia (ii) in front of the retina (iii) concave lens (iv) excessive curvature of the lens; elongation of eyeball	1+1+1+2
	OR: (i) spreading of light by particles (ii) path of light becomes visible in a room through a small hole due to dust (iii) fine molecules scatter blue light more (iv) large particles scatter all colours equally	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 10: **The Human Eye and the Colourful World** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** real and inverted	Convex lens
2	**(a)** ciliary muscles	Accommodation
3	**(d)** convex lens	Converging
4	**(c)** Incorrect — red has the longest wavelength	Violet shortest
5	**(b)** Student A is correct	Sun behind observer
6	**(b)** pupil	Controlled by iris
7	**(b)** retina	Rods and cones
8	**(c)** scattering of light	Molecules
9	**(d)** presbyopia	Both near and far
10	**(c)** violet to red	Violet bends most

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It controls the size of the pupil and hence the amount of light entering the eye.	2
12	To reduce the amount of light entering the eye.	2
13	Scattering of light by colloidal particles, making the path of light visible.	2
14	False — myopia is corrected by a concave lens.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ability of the eye lens to change its focal length to see near and far objects; ciliary muscles weaken and the lens loses flexibility	2+1
16	The focal length of the eye lens cannot be decreased below a certain limit; the image would form behind the retina	3
17	Concave lens diverges rays from distant objects so that they appear to come from the far point of the myopic eye; image formed on the retina	3
	OR: No atmosphere in space — no scattering; on Earth, fine air molecules scatter shorter wavelengths (blue) more	1½+1½
18	Sunlight enters raindrops; refraction and dispersion; internal reflection; refraction on exit; seen in the direction opposite the Sun	3
	OR: Starlight bends towards the normal as it enters denser layers of the atmosphere, so stars appear raised	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) cannot see nearby objects; near point beyond 25 cm (ii) image formed behind retina; convex lens converges rays to form image on retina (iii) long focal length of eye lens; eyeball too small	1+3+1
	OR: (i) twinkling of stars, advanced sunrise/delayed sunset (ii) changing refractive index of the atmosphere (iii) the refractive index of hot air fluctuates	2+2+1
20	(i) presbyopia (ii) weakening of ciliary muscles; lens loses flexibility (iii) bifocal (iv) upper part concave for far vision, lower part convex for near vision (v) LASIK/contact lenses/cataract surgery	1+1+1+1+1
	OR: (i) band of seven colours (ii) different colours travel with different speeds and bend by different angles (iii) two identical prisms, the second inverted (iv) violet	1+2+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.